

Diffusion Prior-Based Amortized Variational Inference for Noisy Inverse Problems

Sojin Lee, Dogyun Park, Inho Kong, Hyunwoo J. Kim

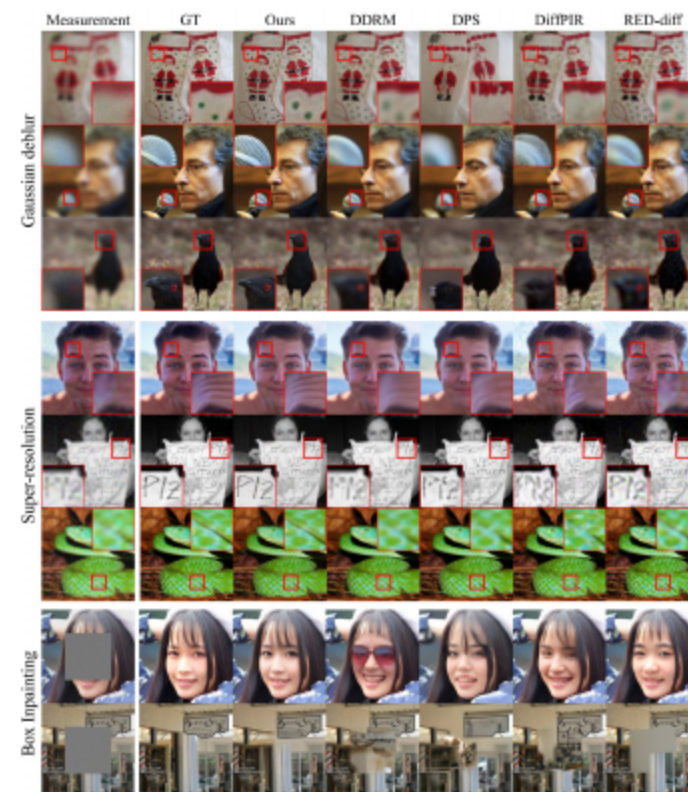
01 Ill-Posed Nature And Computational Bottlenecks



- Inverse problems are ill-posed; many clean signals map to the same noisy measurement.
- Diffusion priors help realism but require long iterative sampling and per-measurement optimization.
- High latency and poor scalability hinder deployment across varying noise levels.
- A single learned mapping could produce posterior samples quickly without knowing exact noise.

01 Target Tasks And Practical Demands

- Target tasks: Gaussian deblurring, 4× super-resolution, and box inpainting with additive noise.
- Applications need real-time or batched processing in imaging and astronomy.
- Desired: fidelity to measurements, perceptual realism, robustness to noise mismatch, minimal evaluations.
- Evaluation uses consistent pretrained diffusion priors across datasets.



02 Related Work And Their Limitations

Diffusion Priors For Inverse Problems

Existing methods modify reverse diffusion via SVD domains, pseudo-inverse, or posterior scores.

Strong quality but rely on long diffusion trajectories and known noise scales.

Performance degrades with noisy pseudo-inverses or noise-scale mismatch.

Variational variants still require multi-step diffusion sampling.

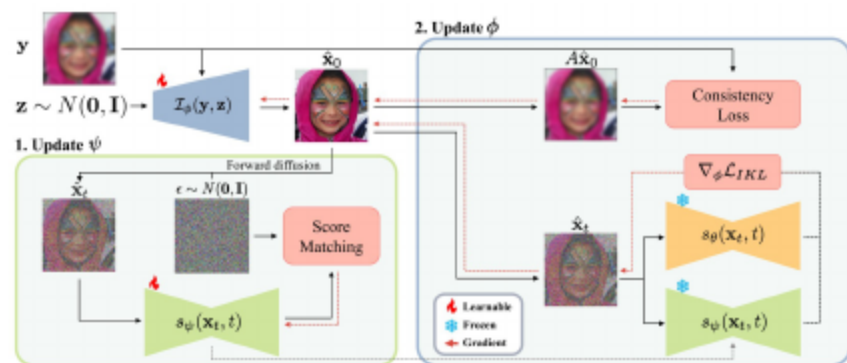
Amortized Variational Inference Basics

Classical VI optimizes per-sample parameters—inefficient and memoryless. Amortized VI maps observations to variational posteriors, enabling zero-shot inference.

Applying AVI with diffusion priors promises single-step sampling and generalization.

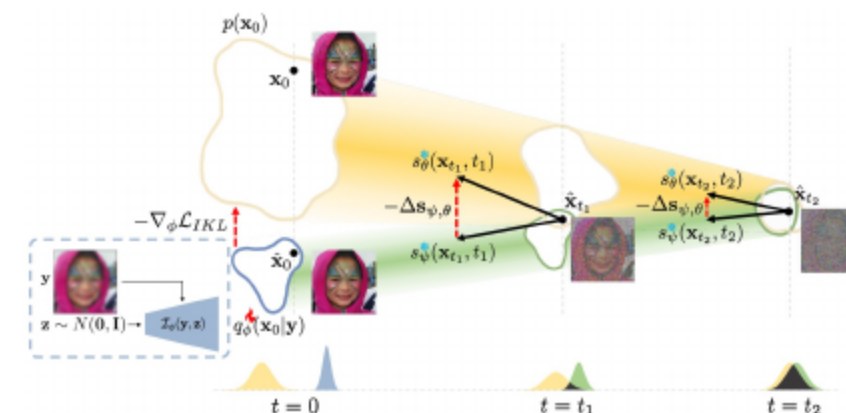
03 Amortized Posterior Mapping With Diffusion Prior

- Introduce DAVI: learns an implicit posterior conditioned on measurements.
- Enables single-step sampling using a pretrained diffusion prior.
- Generalizes to unseen measurements without test-time optimization.

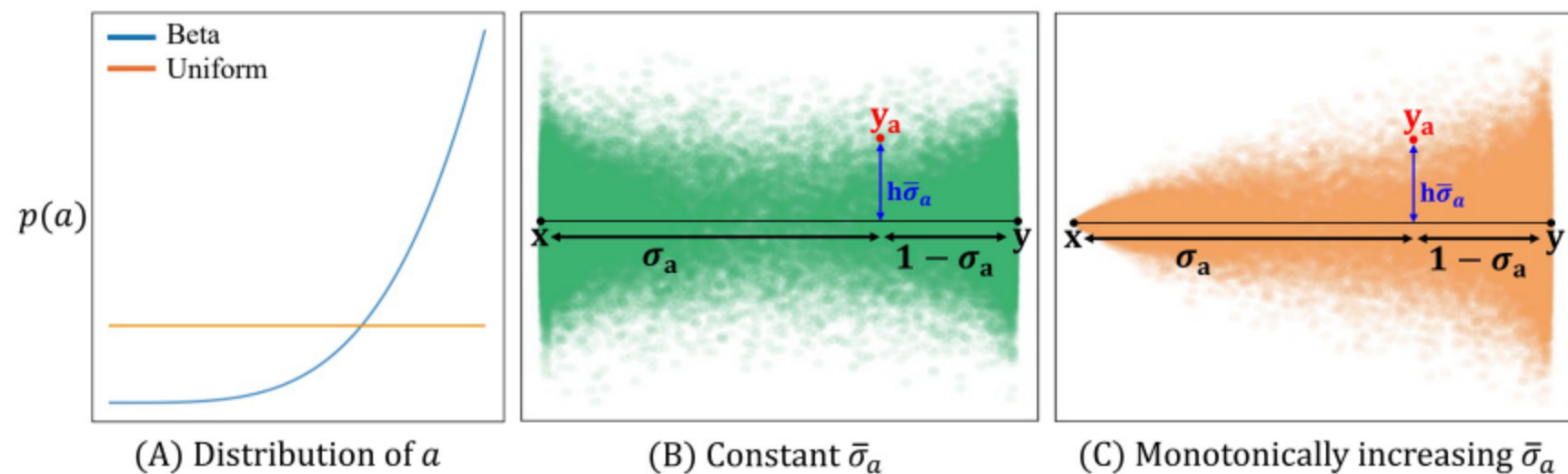


03 Integral KL With Score Distillation

- Minimize Integral KL between implicit posterior and prior marginals.
- Forward SDE smoothing stabilizes training under support mismatch.
- Score-distillation gradient aligns implicit and prior scores.



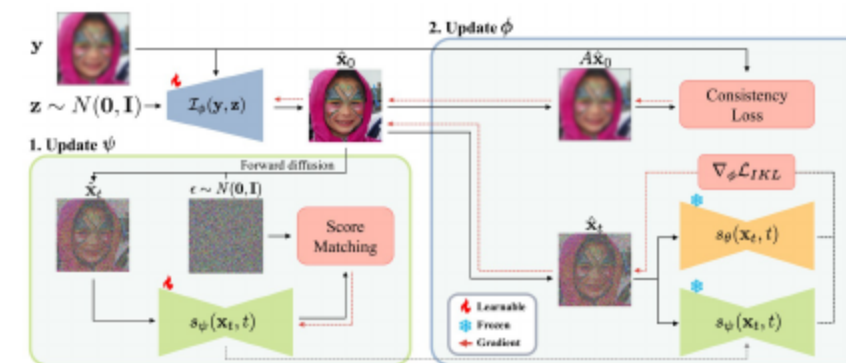
03 Perturbed Posterior Bridge For Generalization



- PPB generates perturbed intermediates between measurement and clean signal.
- Scheduling and noise broaden coverage and training signal.
- Improves robustness and performance under noisy, varied conditions.

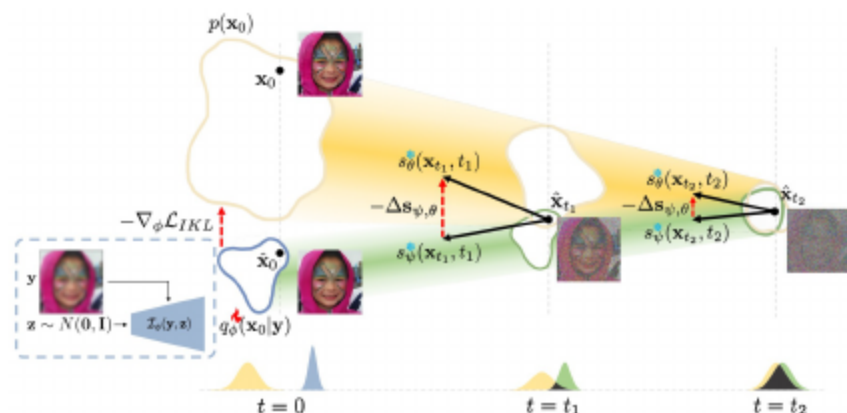
04 Implicit Posterior And Single-Step Sampling

- Implicit network maps measurement and noise to clean sample.
- At test time, single forward pass with $a=1$ yields a posterior sample.
- Eliminates per-measurement optimization and multi-step diffusion.



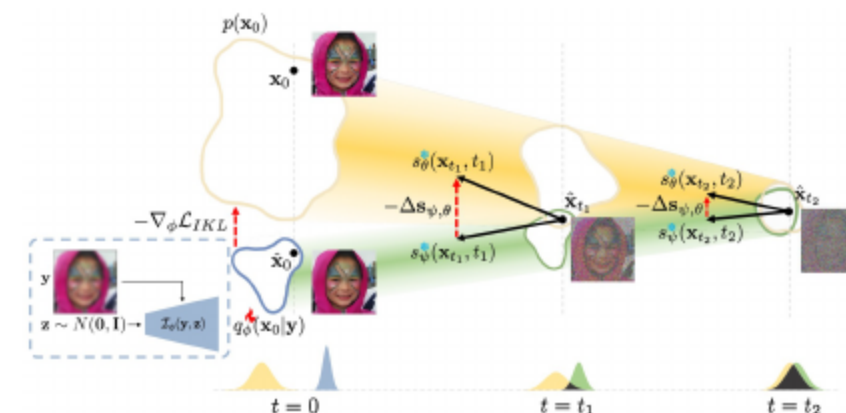
04 Objective: Data Consistency And IKL

- Optimize Gaussian likelihood for measurement consistency.
- Add IKL across diffusion times between implicit and prior marginals.
- Time-integrated KL prevents divergence from support mismatch.

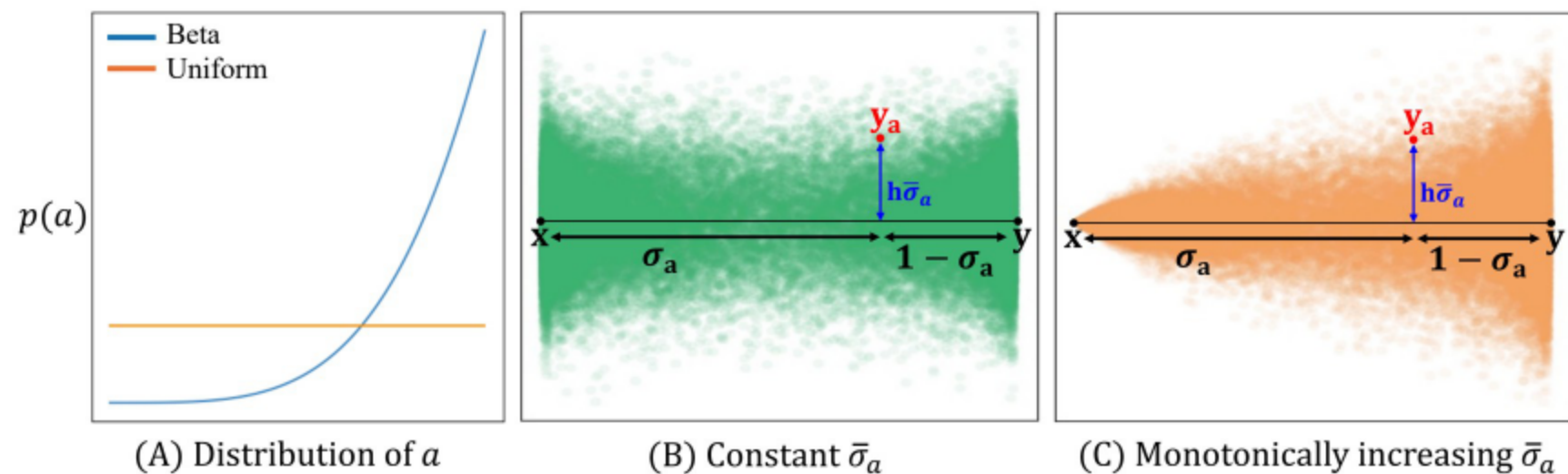


04 Alternating Optimization Strategy

- Train implicit score via denoising score matching to estimate $q_φ$ scores.
- Update $φ$ using score-distillation gradient $s_ψ - s_θ$ plus data consistency.
- Iteratively align implicit posterior with diffusion prior.



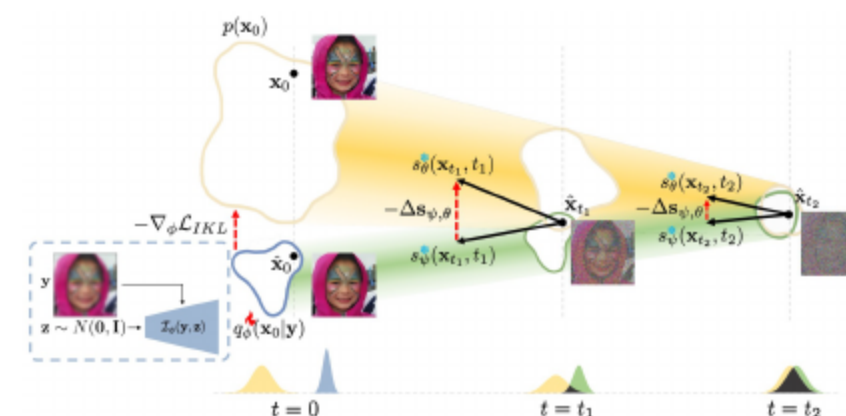
05 Step 1: Sample Data And Build PPB Inputs



- Simulate y from x_0 via Hx_0 with Gaussian noise.
- Sample bridge parameter a and compute σ_a , perturbation scale.
- Form y_a by interpolating y and x_0 with added noise for supervision.

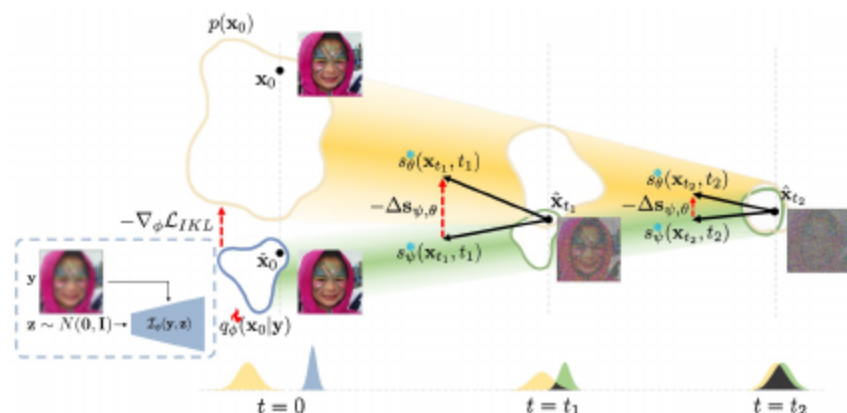
05 Step 2: Update Implicit Score Network

- Sample time t ; diffuse to x_t via forward SDE.
- Train $s_ψ(x_t, t)$ with denoising score matching for $q_φ(x_t|y)$.
- Provides stable gradients for IKL without explicit density.



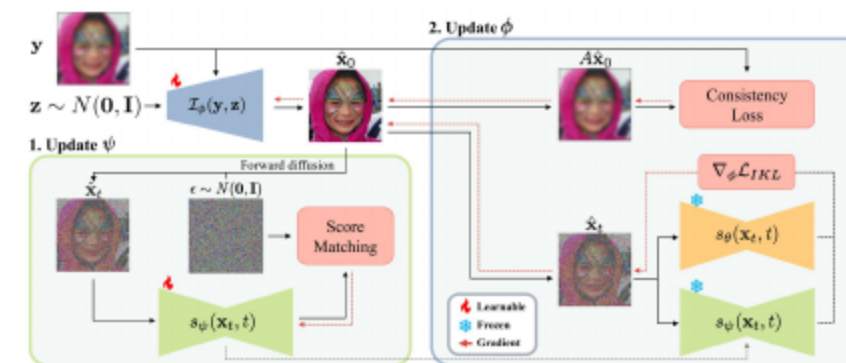
05 Step 3: Update Implicit Posterior Network

- Compute Gaussian likelihood data-consistency loss.
- Estimate IKL gradient using $s\psi - s\theta$ over time.
- Backpropagate through x_t to update ϕ , balancing fidelity and prior alignment.

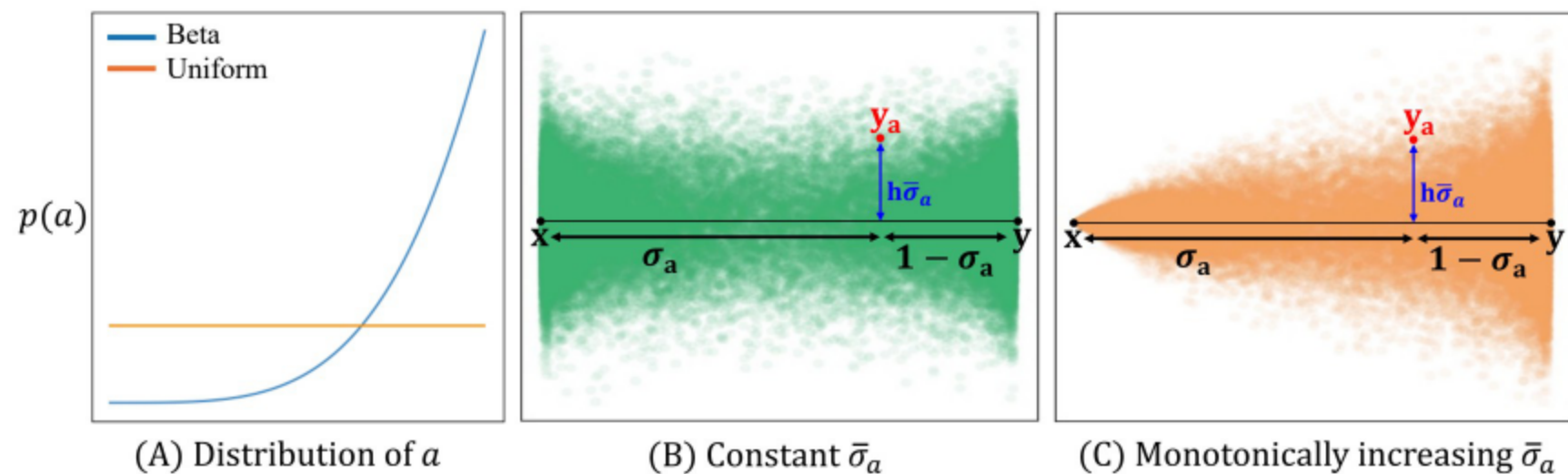


05 Step 4: Single-Step Inference Procedure

- Input y with $a=1$ and small Gaussian z scaled by h .
- Single forward pass of $I\phi$ outputs posterior-consistent reconstruction.
- No diffusion sampling or iterative optimization required.

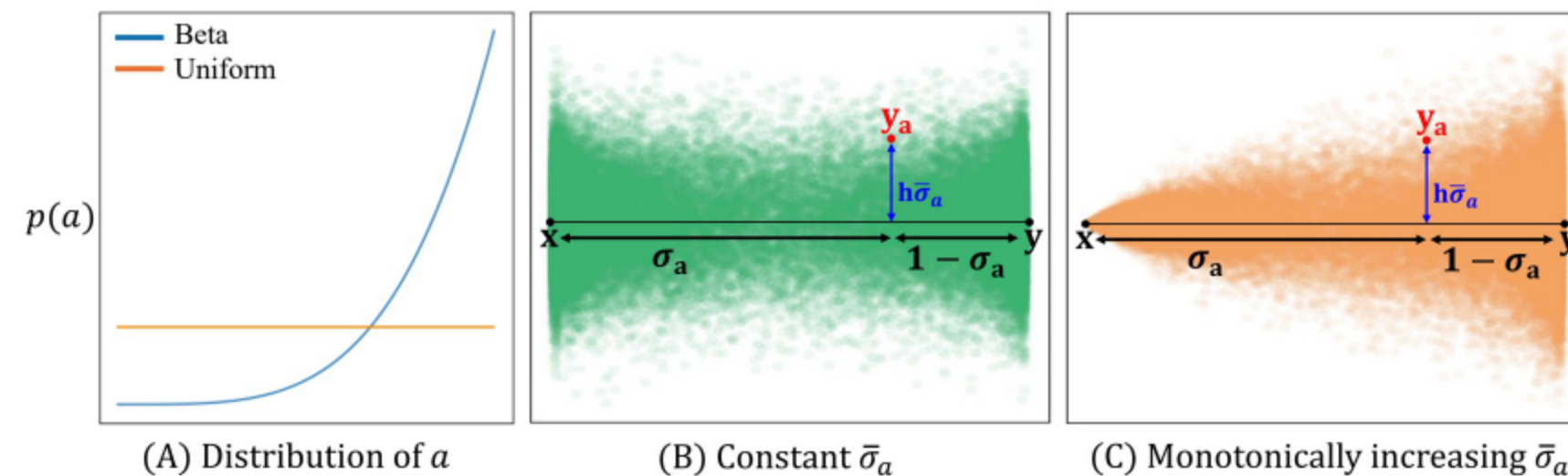


06 Bridge Construction And Scheduling



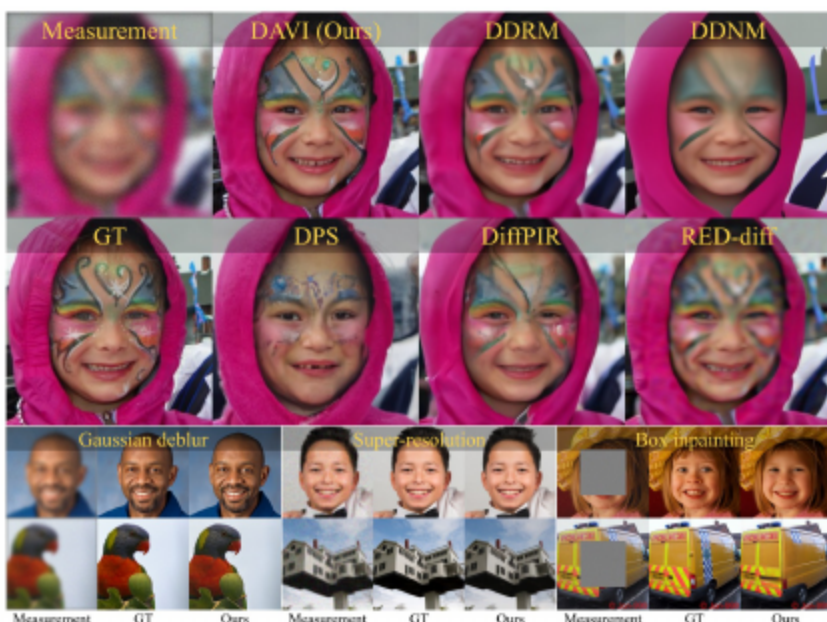
- Construct y_a by interpolating y and x_0 ; add noise via perturbation schedule.
- Sample a (e.g., Beta) and embed into $I\phi$ conditioning.
- Rising perturbation near y improves coverage and generalization.

06 Effect On Generalization And Stability



- PPB broadens training distribution and reduces overfitting to specific degradations.
- Guides $I\phi$ to handle diverse measurement qualities.
- Monotonic perturbation near y with Beta a achieves best fidelity-realism trade-off.

07 Tasks, Data, And Baselines



- Datasets: FFHQ and ImageNet at 256×256 resolution.
- Tasks: Gaussian deblur, 4× SR, and box inpainting with 128×128 masks.
- Baselines: DDIM, DDNM+, DPS, IIGDM, DiffPIR, RED-diff with same prior.

07 Metrics And Efficiency

Method	DAVI (Ours)	RED-diff [23]	DPS [4]	IIGDM [32]	DDRM [18]	DDNM ⁺ [39]
Noise scale σ_y	-	-	-	Required	Required	Required
PSNR $\uparrow$	23.47	<u>23.25</u>	19.81	21.80	16.09	19.25
LPIPS $\downarrow$	0.347	0.470	0.447	<u>0.446</u>	0.595	0.543
FID $\downarrow$	63.37	111.08	<u>64.00</u>	82.98	157.38	147.30

- Report PSNR, LPIPS for fidelity/structure, and FID for realism.
- Track number of function evaluations and wall-clock time.
- DAVI targets superior FID/LPIPS with competitive PSNR at NFE=1.

08 Quantitative Performance Across Tasks

Method	NFE $\downarrow$	Gaussian deblur			4× Super-resolution			Box Inpainting		
		PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
DDRM [18]	20	26.26	<u>0.269</u>	56.82	28.09	0.228	48.19	22.27	<u>0.177</u>	37.62
DDNM ⁺ [39]	100	24.19	0.348	91.48	<u>28.17</u>	0.244	59.09	23.79	0.218	48.82
DPS [4]	1000	21.88	0.315	34.47	25.55	0.255	<u>36.29</u>	21.94	0.302	42.45
IIGDM [32]	100	23.05	0.320	52.72	27.73	<u>0.225</u>	49.86	23.04	0.220	32.99
DiffPIR [45]	100	24.41	0.299	<u>33.91</u>	25.32	0.341	40.35	23.97	0.195	<u>31.27</u>
RED-diff [23]	1000	26.44	0.324	46.55	26.75	0.379	92.82	<u>24.16</u>	0.216	35.80
DAVI (Ours)	1	25.46	0.225	29.92	28.23	0.171	23.96	26.25	0.112	14.31

- DAVI achieves SOTA FID/LPIPS across tasks on FFHQ and ImageNet.
- Often matches or exceeds PSNR of iterative methods.
- Largest FID gains in super-resolution and box inpainting.

08 Robustness To Unknown Noise Scales

Method	DAVI (Ours)	RED-diff [23]	DPS [4]	IIGDM [32]	DDRM [18]	DDNM ⁺ [39]
Noise scale σ_y	-	-	-	Required	Required	Required
PSNR $\uparrow$	23.47	<u>23.25</u>	19.81	21.80	16.09	19.25
LPIPS $\downarrow$	0.347	0.470	0.447	<u>0.446</u>	0.595	0.543
FID $\downarrow$	63.37	111.08	<u>64.00</u>	82.98	157.38	147.30

- Maintains strong LPIPS/FID and competitive PSNR with unknown σ_y .
- Methods requiring σ_y degrade significantly under mismatch.
- Supports practical deployment without noise calibration.

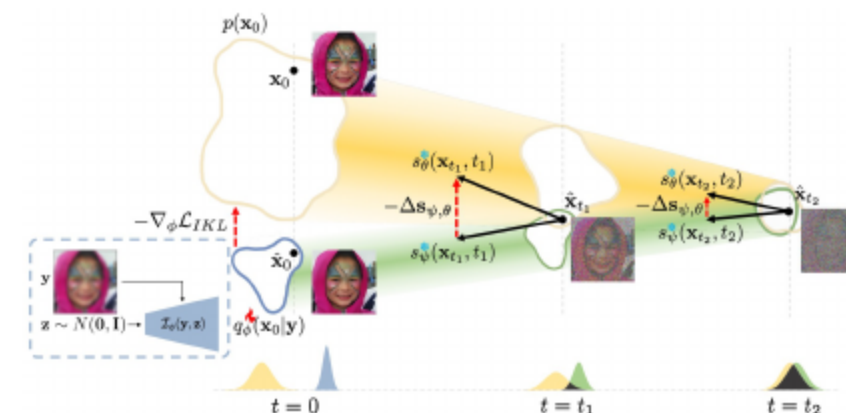
Method	DAVI (Ours)	RED-diff [23]	DPS [4]	ITGDM [32]	DDRM [18]	DDNM ⁺ [39]
Noise scale σ_y	-	-	-	Required	Required	Required
PSNR $\uparrow$	23.47	23.25	19.81	21.80	16.09	19.25
LPIPS $\downarrow$	0.347	0.470	0.447	<u>0.446</u>	0.595	0.543
FID $\downarrow$	63.37	111.08	<u>64.00</u>	82.98	157.38	147.30

- Single-step inference: ~0.04 s/image on single GPU.
- Outperforms fastest iterative baseline with better quality.
- No diffusion trajectories or per-sample optimization needed.

Config.	IKL	PPB	$\bar{\sigma}_a$	h	$p(a)$	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
(A)	X	X	-	-	-	21.47	0.443	50.69
(B)	✓	X	-	-	-	24.70	0.236	37.61
(C)	✓	✓	Const.	0	Uniform	26.71	0.247	49.37
(D)	✓	✓	Const.	0.01	Uniform	<u>26.17</u>	0.239	47.41
(E)	✓	✓	Const.	0.01	Beta	25.99	<u>0.232</u>	<u>34.34</u>
(F)	✓	✓	M.I	0.01	Beta	25.46	0.225	29.92

- PPB with perturbation and Beta-distributed a outperforms linear bridges.
- Monotonically increasing perturbation near y yields best LPIPS/FID.
- Emphasizing challenging intermediates improves generalization.

- Removing IKL markedly worsens realism.
- Forward SDE smoothing prevents support mismatch pathologies.
- IKL aligns q_ϕ with prior across times, improving FID.



- DAVI reframes diffusion-prior inverse problems as amortized VI for single-step sampling.
- Achieves strong realism and fidelity, robust to noise-scale uncertainty.
- PPB and IKL are key for stability and generalization.
- Future: broader degradations, new sensing modalities, ethical safeguards.

The image features a minimalist design on a white background. A central white parallelogram with a dark green border contains the text 'THANKS!'. This parallelogram is flanked by two solid green rectangles. Various green geometric shapes, including triangles and lines, are scattered around the central elements. A thin black horizontal line runs across the middle of the image, and a thin black vertical line runs down the right side, intersecting at the center.

THANKS!